AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15/Nov./2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2h

Instructions:

- Total Marks: **60**
- This exam consists of **6** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question (1) Choose the correct answer (10 Marks): -

1. Which of the following is a key characteristic of classical encryption techniques?
   a. Use of asymmetric keys
   b. Lack of complexity in algorithms
   c. Use of substitution and transposition methods
   d. Dependence on modern computational power
2. In number theory, what is the greatest common divisor (GCD) of two integers?
   a. The smallest prime factor
   b. The largest number that divides both integers without leaving a remainder
   c. The sum of the two integers
   d. The product of the two integers
3. The Data Encryption Standard (DES) operates on what block size?
   a. 32 bits
   b. 64 bits
   c. 128 bits
   d. 256 bits
4. Which of the following describes the Advanced Encryption Standard (AES)?
   a. A block cipher that operates on 64-bit blocks
   b. A symmetric encryption algorithm with variable key lengths of 128, 192, or 256 bits
   c. An asymmetric encryption algorithm
   d. A form of digital signature
5. Which operation is NOT part of the block cipher mode of operation?
   a. Cipher Block Chaining (CBC)
   b. Electronic Codebook (ECB)
   c. Hash Function
   d. Counter (CTR)
6. In the context of stream ciphers, what is the main purpose of a key stream?
   a. To provide authentication
   b. To generate a fixed block length
   c. To encrypt plaintext in real-time
   d. To ensure data integrity
7. Which of the following describes an important aspect of network security?
   a. Only data encryption
   b. Maintaining data confidentiality, integrity, and availability
   c. Regular software updates
   d. Physical security measures
8. What is a common method of generating random bits for encryption systems?
   a. Using a predetermined algorithm
   b. Hardware random number generators
   c. Simple toggling of a switch
   d. Human-generated input
9. In asymmetric encryption, what is the role of the public key?
   a. It is kept secret and used for decryption
   b. It is shared openly and used for encryption
   c. It is used to sign messages
   d. It is irrelevant in modern communication
10. Which of the following best characterizes the use of round functions in block ciphers?
    a. They provide non-reversibility
    b. They ensure a 32-bit output
    c. They enhance security through confusion and diffusion
    d. They are used only in symmetric encryption

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Question (2) Answer the following questions (10 Marks): -

1. Explain the primary purpose of encryption in network security?(2 Marks)
2. What is a substitution cipher?(2 Marks)
3. Explain the significance of the P-box in a block cipher.(2 Marks)
4. Decrypt the Playfair message "KXJEY UREBE ZWEHE WRYTU HEYF" using key "royal new zealand navy". (4 Marks)

Question (3) Answer the following questions (10 Marks): -

1. Explain the concept of confusion in cryptography.(2 Marks)
2. Explain the role does the S-box play in the AES algorithm? (2 Marks)
3. How does the Feistel structure enhance the security of block ciphers? (2 Marks)
4. Decrypt the Caesar ciphertext "MJQQT" (shift unknown, 0-25) by testing shifts to find meaningful English. Identify the shift and plaintext.(4 Marks)

Question (4) Answer the following questions (10 Marks): -

1. What function does the key schedule serve in the AES algorithm? (2 marks)
2. What is the difference between a stream cipher and a block cipher? (2 marks)
3. Describe the purpose of initialization vectors (IVs) in block cipher modes like CBC. (2 marks)
4. The ciphertext "WKHXUHQZ" (from a monoalphabetic substitution cipher) has letter frequencies: W=2, K=2, H=2, X=1, U=1, R=1, Q=1, Z=1. English frequencies: E=12%, T=9%, A=8%. Map the most frequent ciphertext letter to 'E' and the second to 'T'. Decrypt assuming it's "THEQUICK".(4 Marks)

Question (5) Answer the following questions (10 Marks): -

1. What is the purpose of the Round Function in AES? (2 marks)
2. Describe one method to attack stream ciphers. (2 marks)
3. What are the differences between symmetric and asymmetric encryption? (2 marks)
4. Using the extended Euclidean algorithm, find the multiplicative inverse of 567 modulo 890(2 Marks)
5. Decrypt the ciphertext "WACAWETTKDANATDN" using the 3-rail fence cipher (2 Marks)

Question (6) Answer the following questions (10 Marks): -

1. What is meant by "data integrity" in networking? (2 Marks)
2. What is the difference between a stream cipher and a block cipher? (2 Marks)
3. What is the significance of the key length in symmetric encryption algorithms? (2 Marks)
4. Calculate the following: (2 Marks)
   a. $\varphi(420)$ *b.* $\varphi(504)$ *c.* $\varphi(385)$ *d.* $\varphi(720)$
5. A stream cipher uses keystream bits 10110110 XORed with plaintext bits 11001001 to produce ciphertext 01111111. Recover the plaintext (2 Marks)

End of Questions

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